

STRATEGIC BLUEPRINT: PILLAR 3

Financial Security, Value-Based ROCI & Revenue Cycle Intelligence

PROGRAM: Psychiatric Hospital Analytics Architecture

PHASE: Phase 5 — Cross-Domain Longitudinal Evidence & Financial Intelligence Synthesis

TARGET PILLAR: PILLAR 3 — Financial Security & Revenue Cycle Management

EXECUTIVE SUMMARY & CROSS-PILLAR INTEGRATION MATRIX

Pillar 3 establishes the financial intelligence layer of the Warsaw Psychiatric Hospital Analytics Program. Rather than evaluating billing and claim denials in isolation, Pillar 3 integrates verified empirical findings from **Pillar 1 (Clinical Outcomes & Care Continuity)** and **Pillar 2 (Operational Efficiency & Capacity Optimization)** to evaluate healthcare delivery through a **Value-Based Return on Clinical Investment (ROCI)** framework.

PILLAR 3 MASTER MODULE ARCHITECTURE

The 4 core analytical inquiry domains of Pillar 3 are designed to isolate administrative friction, quantify treatment cost-efficacy, prevent audit liabilities, and optimize payer SLA contracts.

PILLAR 3 MASTER INQUIRY MODULE DOMAINS	
MODULE 3.1 (ANA-301): Prior Authorization Latency, Specialty Capacity & Claim Denial Cascades	
MODULE 3.2 (ANA-302): Value-Based ROCI: Cost-Efficacy of Regimens vs. Digital Engagement Buffer	
MODULE 3.3 (ANA-303): Audit Vulnerability, Provider Handoffs & Clinical Note Documentation Gaps	
MODULE 3.4 (ANA-304): Acute Crisis Financial Hemorrhage & Payer Contract SLA Optimization	

MODULE 3.1: Prior Authorization Latency, Specialty Capacity & Claim Denial Cascades

- Module ID: ANA-301 (Inquiry Domain 3.1)
- Target Schema Anchors: dbo.Financials (TBL-007), dbo.Insurance_Workflow (TBL-008), dbo.Appointments (TBL-001), dbo.Providers (TBL-014), dbo.Treatments (TBL-017)
- Analytical Target Question:

How do mandatory prior authorization requirements (Prior_Authorization_Required) and approval latencies (Authorization_Delay_Days) intersect with provider specialties (Specialty) and scheduling delays (Wait_Time_Days) to drive insurer claim rejections (Claim_Rejected_Flag = 1) and direct unrecovered billed revenue loss (Total_Billed_PLN)?

- Analytical Causality Mechanics (The "Why"):

Complex prior authorization requirements imposed by payers introduce administrative friction. When approval latencies (Authorization_Delay_Days ≥ 14) compound existing operational scheduling backlogs (Wait_Time_Days ≥ 30), treatment windows

expire or billing codes mismatch. This triggers formal claim denials from third-party payers, directly converting operational latency into unrecoverable financial write-offs.

- **Observed Output Pattern & Impact:**

Isolates specific payer types (Insurance_Type) and clinical specialties where prior authorization processes fail to offer financial protection (as proven in P2 where authorized cohorts still generated 116.74M PLN in financial leakage). Equips finance executives to automate point-of-scheduling eligibility verification and renegotiate payer SLAs.

- **Future Strategic Recommendations Generated:**

1. Overhaul payer SLA contracts by establishing mandatory 48-hour prior authorization turnaround windows.
2. Implement automated pre-submission claim scrubbers for high-volume specialties (Addiction Specialist, Psychiatry).

MODULE 3.2: Value-Based ROCI: Cost-Efficacy of Regimens vs. Digital Engagement Buffer

- **Module ID:** ANA-302 (Inquiry Domain 3.2)

- **Target Schema Anchors:** dbo.Financials (TBL-007), dbo.Treatments (TBL-017), dbo.Longitudinal_Sessions (TBL-009), dbo.Outcomes (TBL-011), dbo.Medication_History (TBL-010)

- **Analytical Target Question:**

What is the structural distribution of total billed revenue (Total_Billed_PLN) across combination therapy modalities (Therapy_Type + Medication) relative to observed clinical trajectories (Trajectory_Outcome), and how does digital/behavioral engagement (App_Usage_Hours, Homework_Completed) modify the financial cost per unit of clinical improvement?

- **Analytical Causality Mechanics (The "Why"):**

High-cost pharmacological regimens with frequent medication alterations and severe side effects do not inherently yield superior clinical recovery (as established in P1 where net clinical improvement dropped by -8.78). Conversely, low-cost behavioral self-management tools (Homework_Completed = 1, App_Usage_Hours > 8.5) stabilize patient trajectories (74.76% remission/improvement rate). Quantifying the cost per successful remission establishes a value-based healthcare baseline.

- **Observed Output Pattern & Impact:**

Identifies high-cost, low-efficacy therapeutic protocols and proves the financial return on clinical investment (ROCI) of integrating digital mental health platforms.

- **Future Strategic Recommendations Generated:**

1. Shift clinical protocols toward high-ROCI care pathways that combine targeted pharmacology with mandatory digital self-management.
2. Reallocate hospital capital investments from low-yield combination therapies toward digital engagement infrastructure.

MODULE 3.3: Audit Vulnerability, Provider Handoffs & Clinical Note Documentation Gaps

- **Module ID:** ANA-303 (Inquiry Domain 3.3)
- **Target Schema Anchors:** dbo.Insurance_Workflow (TBL-008), dbo.Financials (TBL-007), dbo.Provider_Assignment_History (TBL-013), dbo.Clinical_Notes (TBL-004), dbo.Treatments (TBL-017)

- **Analytical Target Question:**

How do high provider reassignment rates (Changed_Provider_Flag = 1) driven by clinician burnout and negative qualitative documentation sentiment (Sentiment_Score < -0.40) correlate with insurance appeals (Appeal_Required), partial claim payments (Partial_Payment_Flag), and fraud investigation flags (Fraud_Investigation_Flag) among high-billed treatments?

- **Analytical Causality Mechanics (The "Why"):**

Overloaded clinicians experiencing severe burnout (70.8% operating at > 40 cases/week) experience elevated absenteeism (15.95 sick days/year), forcing frequent patient handoffs (17.91% reassignment rate). These handoffs disrupt therapeutic rapport and create gaps in unstructured clinical notes (Clinical_Notes). Third-party payers audit these high-cost treatments; lacking continuous, high-quality documentation, payers enforce partial payment deductions or initiate fraud investigations.

- **Observed Output Pattern & Impact:**

Exposes the hidden billing compliance liability caused by clinical workforce distress, protecting the hospital against retroactive insurer clawbacks and write-offs.

- **Future Strategic Recommendations Generated:**

1. Enforce mandatory clinical documentation templates during provider handoffs to ensure continuous medical necessity justification.
2. Deploy automated AI documentation scrubbers to flag missing narrative justification prior to billing submission.

MODULE 3.4: Acute Crisis Financial Hemorrhage & Payer Contract SLA Optimization

- **Module ID:** ANA-304 (Inquiry Domain 3.4)

- **Target Schema Anchors:** dbo.Financials (TBL-007), dbo.Crisis_Events (TBL-006), dbo.Appointments (TBL-001), dbo.Patients (TBL-012), dbo.Comorbidities (TBL-005)

- **Analytical Target Question:**

What is the total financial volume (Total_Billed_PLN) consumed by emergency room visits (ER_Visit_Flag) and acute hospitalizations (Hospitalization_Flag) resulting from scheduling delays (Wait_Time_Days $\geq$ 30) and non-attendance (Status = 'No-Show'), and how does this financial drain distribute across diagnostic profiles (Primary_Diagnosis) and payer groups (Insurance_Type)?

- **Analytical Causality Mechanics (The "Why"):**

When patients facing severe mental illness encounter extended scheduling latencies (54.26% of visits wait an average of 49.53 days) or accessibility barriers (11.08% no-show rate in remote districts), their conditions acutely decompensate. This forces emergency department utilization (8,428 crisis events, 4,228 hospitalizations), transferring low-cost outpatient care into high-cost acute inpatient stays that face elevated claim rejection risks.

- **Observed Output Pattern & Impact:**

Quantifies the exact monetary loss caused by outpatient access bottlenecks and equips executive leadership with empirical data to negotiate risk-sharing or capitated payment models with payers.

- **Future Strategic Recommendations Generated:**

1. Establish rapid-access crisis diversion clinics for high-risk diagnostic profiles to prevent emergency hospitalizations.
2. Implement value-based shared-savings contracts with payers that penalize scheduling delays and reward crisis prevention.